# PatientAdvocateLM: A Clinically Grounded Retrieval-Augmented Conversational Agent for Medical Visit Preparation

Anonymous ACL submission


## Abstract

We explore an LLM-driven patient advocate to help patients clarify concerns and ask questions for their next medical visits. Given the high-stakes, patient-facing, and open-ended nature of the task, our study is highly exploratory in both design and evaluation. We explore designs that structure free-form LLM interaction with a clinically validated framework and enhance information quality by retrieving content from vetted sources. We evaluate our system against a vanilla LLM through a within-subject study with ten human patients in the context of localized prostate cancer, assessing cognitive load during preparation. We also perform an evaluation with 43 simulated patients using reference-based metrics. Results from both evaluations complementarily show that our system can effectively help patients prepare for their medical visits. Compared to a vanilla best-performing LLM, our system generates more relevant questions and reduces patients' cognitive load during preparation. We also identify several design lessons to guide future refinement.


## 1 Introduction

Many medical conditions are complex and require shared decision-making between patients and clinicians beyond clinical standards, yet limited appointment time and passive patient engagement can hinder effective communication (Martinez-Gonzalez et al., 2019). Examples include chronic conditions like eczema with various management strategies, low-risk cancers with multiple treatment options, and other scenarios involving trade-offs between efficacy, side effects, and quality of life. Helping patients articulate questions, clarify preferences, and actively participate is crucial for strong bidirectional patient-provider communication (Martin and Tipton, 2007). Traditionally, Patient Advocates have fulfilled this role, helping patients' interests and needs to be heard (Abbasinia et al., 2020; Martin and Tipton, 2007). However, current healthcare financing structures do not support their widespread adoption, leaving patients to navigate these communication challenges largely alone (Negarandeh et al., 2006; Nsiah et al., 2020).

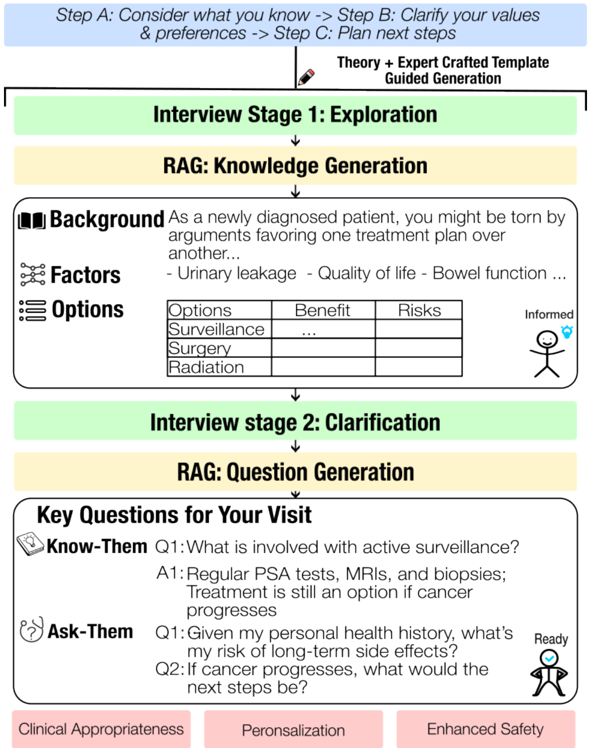


Figure 1: Design ideas of an agent for medical visit preparation: integrating Ottawa Decision Guide (a shared decision-making framework) with retrieval-augmented generation into conversations.

Motivated to build an intelligent assistant that, like a traditional patient advocate, centers on patients' concerns and needs, we propose developing a Large Language Model (LLM)-driven patient advocate, focusing on one specific advocacy function: helping patients prepare for upcoming clinical visits. This task presents unique challenges despite existing medical NLP applications: it requires addressing patient knowledge gaps, exploring personal values and preferences, and carefully suggesting a visit plan while balancing clinical appropriateness with responsiveness, all within an open-ended, patient-centered context without clear ground truth. Given its high-stakes, novel application nature, our study is highly exploratory, manifesting in both system design and evaluation approaches.

This paper makes three contributions: **First**, We present (to our knowledge) an early attempt to apply LLMs in such high-stakes, patient-facing scenario – medical visit preparation, and detail our exploratory designs that balance personalization with controllability, safety, and clinical appropriateness. Our exploration includes using a clinically validated framework, the Ottawa Personal Decision Guide, to structure an LLM-assisted interview that helps patients progress from being uninformed to visit-ready, and integrating retrieval-augmented generation (RAG) to enhance information safety by grounding responses on vetted sources. **Second**, We explore evaluation methods to investigate how this design differs from a basic LLM used by patients. We conduct a small-scale study with 10 real patients in localized prostate cancer, using cognitive load assessment to examine whether our design reduces mental effort during visit preparation. To explore larger-scale testing, we build 43 simulated patients from online forum posts and develop reference-based evaluation to indicate the system's ability to generate patient highly concerned questions. Both evaluations complement each other, providing insights into the benefits of our design from multiple perspectives. **Lastly**, We provide empirical findings and design lessons that reveal where the designs provide value over plain LLMs, and identify key limitations and requirements for safe deployment of patient-facing AI systems in clinical settings.

## 2 Related Work

Recent LLM deployment in healthcare has focused mainly on augmenting clinician capabilities (Tu et al., 2024; Wu et al., 2025; Kim et al., 2024; Li et al., 2024). One line of work targets diagnostic decision support (e.g., MEDIQ (Li et al., 2024), AMIE (Tu et al., 2024)), while another targets clinical preparation (e.g., PCP-Bot (Amogh Ananda Rao et al., 2025), LLM-CDS (Lajmi et al., 2026)). However, these systems primarily serve providers rather than patients, contrasting with our focus on patient advocacy, specifically helping patients prepare for their visits. Our research problem presents additional challenges. First, our task is more complex: it must address patient knowledge gaps, explore patient needs, and balance clinical appropriateness, safety, and responsiveness. Second, existing tasks are typically more deterministic with ground truth (e.g., diagnosis), making evaluation more feasible (Yu et al., 2025; Hicke et al., 2026). In contrast, visit preparation involves holistic measurement including hard-to-measure mental effort and open-ended question generation with many potentially valid options.

## 3 Methodology

### 3.1 Task Definition

To enable systematic exploration of this task, we first formalize it as follows. Given a disease type $D$ and a patient concern type $T \in \mathcal{T}$ (where $\mathcal{T}$ is the set of predefined concern categories), an agent conducts a multi-turn conversation with a patient to generate a conversation $C = \{(a_i, p_i)\}_{i=1}^{n}$, where $a_i$ and $p_i$ represent the agent's and patient's utterances at turn $i$, respectively. During the conversation, the agent interacts with the patient to understand their situation and concerns. The conversation proceeds within a predefined token budget. Upon completion, the agent generates a visit preparation panel containing a question set $Q = Q_{\text{know}} \cup Q_{\text{ask}}$, where $Q_{\text{know}} = \{q_1^{\text{know}}, \ldots, q_5^{\text{know}}\}$ consists of five "know-them" questions (questions answerable using validated sources, optionally paired with answers), and $Q_{\text{ask}} = \{q_1^{\text{ask}}, \ldots, q_5^{\text{ask}}\}$ consists of five "ask-them" questions (questions for the patient to bring to their doctor that cannot be directly answered from validated sources).

### 3.2 System Overview

We design a system to address this task by bridging knowledge gaps, clarifying personal values, and generating personalized questions through a structured workflow grounded in a clinically validated framework and retrieval-augmented generation. The workflow consists of four components, shown in Fig 1, alternating between interview stages and RAG steps: (i) INTERVIEW STAGE 1: explores what the patient already knows about a decision concern; (ii) RAG: KNOWLEDGE GENERATION generates a knowledge panel; (iii) INTERVIEW STAGE 2: clarifies personal values and preferences by reviewing options with their risks, benefits, and certainty levels; and (iv) RAG: QUESTION GENERATION generates visit preparation questions. The interview stages are adapted from the Ottawa Personal Decision Guide framework and guided by expert-crafted template questions, enhanced with LLM-generated flexibility to balance clinical appropriateness with responsiveness (Sec 3.3). The RAG steps retrieve information from validated sources to generate knowledge panels and visit preparation questions, ensuring information quality and safety through controlled retrieval (Sec 3.4).

### 3.3 Framework-Guided Interview Structure

**Theoretical Foundation.** Our interview is adapted from the Ottawa Personal Decision Guide, a validated framework for patients to use in shared decision-making. The OPDG outlines three steps to help patients make informed, value-aligned decisions:

STEP A Consider what the patient already knows about the decision;

STEP B Clarify personal values and preferences by reviewing options along with their risks, benefits, and levels of certainty;

STEP C Identify support needs and plan the next steps.

Originally designed as a largely blank guided worksheet,[1] the OPDG assumes patients can obtain (or already have) the medical knowledge needed to complete it. However, this is not easy. This leaves an ideal opportunity to leverage NLP tools to explore and retrieve information, bridging this gap for patients.

**Expert-Crafted Template Questions.** We work with a clinician from our medical school to transform this static form into an interactive conversation, enabling the agent to probe the patient's situation for Steps A and B respectively: (1) surface the decision landscape, and (2) clarify personal values and concerns. At the first stage, we categorize concern types into 8 categories: "Diagnosis and Screening", "Treatment Plan", "Physical Health Wellness", "Emotional and Mental Health", "Long-Term Management and Monitoring", "Insurance and Financial Support", "Nutrition and Dietary Guidelines", and "Other Concerns". For each category, the expert crafts two interview questions to explore the patient's situation. At the second stage, we craft seven questions to ask about the patient's feelings on options, unclear points, trade-offs, and family support. Each interview involves a set of 9 template questions based on the concern category.

**Template-Guided Response Generation.** Based on early testing feedback (see the first design lesson in Sec 6), we implement a template-guided response generation mechanism that balances responsiveness with framework control. For each turn, the LLM generates a response that: (1) must include the template question exactly as provided, (2) may add one personalized segment before the template question that responds to what the patient just said (acknowledging, providing useful information, or building on their answer), and (3) may optionally add one additional personalized question either before or after the template question. This approach retains personalization while maintaining strict adherence to the clinical structure. Additionally, before the end of each stage, we allow the LLM to freely ask two questions without template constraints, providing further flexibility to explore patient-specific concerns that may not be covered by the structured questions. As a result, each interview typically involves around 13 turns. See case conversation in Appx 4.

[1] OPDG worksheet: https://decisionaid.ohri.ca/docs/das/OPDG.pdf

### 3.4 Retrieval-Augmented Generation

As mentioned earlier, RAG is used at the end of each stage. We detail RAG's two functions: knowledge panel generation after the first stage and visit preparation question generation after the second, exemplified in Fig 1. Note that at these RAG steps, to ensure safety and reduce hallucination, we limit the LLM's creative generation, requiring it to only paraphrase and organize retrieved content.

**Knowledge Panel Generation.** Once the agent has explored the patient's situation, we generate a knowledge to help patients understand the decision they face and key facts related to possible choices. The retrieved content contains: (1) background knowledge about the decision topic, (2) key factors related to the patient's concern, and (3) possible options with their associated benefits and risks, formatted as a grid. The system tracks and cites page numbers from the source material to enhance transparency and verifiability.

**Visit Preparation Question Generation.** Upon confirmation of values and priorities, the second RAG module generates two sets of visit preparation questions:

- *Know-Them Questions:* Questions relevant to the patient's concerns (based on their interview discussions) that are answerable with information from validated patient educational guidebooks, presented along with short answers retrieved directly from the guidebooks.
- *Ask-Them Questions:* Questions personalized to the patient but not directly answerable from the validated guidebooks. Because these questions may be very specific, they are potentially high-

value for the patient to ask at their upcoming clinical visit.

This final step supports patient confidence in communicating with their provider during the clinical visit.

## 4 Evaluation with Human Patients

While patients could use a plain LLM like ChatGPT for visit preparation, unconstrained free-form interaction may lack clinical structure and grounding in validated sources. We evaluate whether our clinically guided, retrieval-augmented framework improves medical visit preparation compared to unconstrained free-form interaction under the same interaction budget and base LLM.

### 4.1 Comparison Setup

**Comparison Approach.** We compare our system against a free-form baseline: a plain LLM with a simple system prompt instructing it to interview a patient for visit preparation. Our system follows a fixed number of turns due to the semi-structured interview design, while the plain LLM may generate longer responses per turn (e.g., asking 10 questions per message).

**Fair Comparison Constraints.** To ensure fair comparison, we constrain both systems to the same token budget, the same base model (GPT-4o), and require outputs in the same form. We estimate our system's average output length over 5 rounds of testing and set this average (1050 tokens) as the shared token budget. Our system always completes its predefined turns, while the plain LLM is instructed to conclude the interview briefly when approaching the budget. After the interview stage, both systems generate five "know-them" and five "ask-them" questions. However, the plain LLM leaves answers to "know-them" questions blank due to lack of knowledge sources. To prevent the plain LLM from offering visit prep suggestions during the interview, we constrain it to only ask exploration questions and provide information, ensuring both systems only recommend visit prep questions at the end.

### 4.2 Procedure and Participants

**Study Procedure.** To ensure the evaluation is focused and feasible, the study is limited to patients with localized prostate cancer, a low-risk cancer with complex medical tradeoffs in decision-making. This disease is selected also because a clinician on our team help customize the templates and select patient guidebooks for this condition. Using a within-subject design, each patient interacts with both systems in random order. The within-subject design was chosen because our measurements focus on cognitive experience with ordinal ratings that are difficult to calibrate across subjects. We allow patients to use different concerns for each system to avoid the first experience influencing the second (e.g., patients becoming better at articulating the same concern after the first system). Both systems use identical interface layouts. The study was conducted with IRB approval.

**Metrics.** We assess cognitive load during visit preparation using a framework adapted from prior work (Klepsch et al., 2017; Chen et al., 2025). Unlike general instruments like NASA-TLX (Hart, 2006) that include irrelevant dimensions (e.g., physical load), this framework focuses on three types of cognitive load: *intrinsic load* (inherent complexity of visit preparation with this concern), *extraneous load* (unnecessary cognitive effort caused by poor system design), and *germane load* (beneficial cognitive effort that supports understanding and reflection). We measure the following metrics: (1) **Overall Mental Effort** on a 1-9 scale, (2) **IL1** (intrinsic load): "The medical issues I needed to think through for this visit were inherently complex (regardless of the system)," (3) **EL1** (extraneous load): "I needed extra time to look for information elsewhere because the system didn't provide enough," (4) **EL2** (extraneous load): "I may need external research to validate the information as I don't know the source of the information," (5) **GL1** (germane load): "The system explores my concerns and provides information in an organized way," and (6) **GL2** (germane load): "The system helped me learn new information or better understand my medical situation and options." All cognitive load questions use 1-7 scales.

**Participants.** We recruited 10 male participants with localized prostate cancer from multiple sources: seven from our university's health research center and the others from Prolific, an online research participant platform[2]. Participants were compensated at 20 US dollars per hour. Inclusion criteria required participants to: (1) be biologically male at birth and over 30 years of age, (2) currently have prostate cancer, and (3) have been scheduled for a relevant medical visit prior to our study.

[2] https://www.prolific.com

### 4.3 Results

Tab 1 presents cognitive load results comparing the plain LLM and our system across 10 human participants. Our system achieves lower mental effort (3.40 vs. 4.30, $p = 0.204$), even when intrinsic load is slightly higher. This is particularly notable because, in our within-subject design, participants discuss different concerns with each system, causing intrinsic load to vary across concerns and serving as a control variable. The improvement in mental effort appears primarily driven by a significant increase in germane load GL1 (6.10 vs. 4.90, $p = 0.018$), indicating that our system's structured support helps reduce cognitive effort. While other dimensions show positive trends favoring our system, the limited sample size prevents reaching statistical significance. See case conversation in Appx 4.

| Metric | Plain LLM | Ours |
|---|---|---|
| Overall Mental Effort ↓ | 4.30 | **3.40** |
| Intrinsic Load (IL1) | 3.30 | 3.70 |
| Extraneous Load (EL1) ↓ | 3.50 | **3.20** |
| Extraneous Load (EL2) ↓ | 3.00 | **2.70** |
| Germane Load (GL1) ↑ | 4.90 | **6.10**$^*$ |
| Germane Load (GL2) ↑ | 5.20 | **5.60** |

Table 1: Cognitive Load Assessment Results (n=10). ↓ indicates lower is better; ↑ indicates higher is better. $^*$ p<0.05 (paired t-test/Wilcoxon test).

## 5 Evaluation with Simulated Patients

Despite initial insights, an evaluation with only 10 human participants limits our ability to draw robust conclusions. However, in the medical domain, recruiting patients for interactive studies is inherently challenging. For a larger-scale evaluation, we explore the use of simulated patients from real patient posts, followed by comparing our system's generated questions against patients' actual questions to evaluate alignment with their information needs.

### 5.1 Comparison Setup

We conduct an evaluation where each simulated patient interacts with both a plain LLM and our designed system. Each simulated patient completes two interview sessions, one with each system, and both systems generate "ask-them" questions (questions recommended for the patient to bring to their doctor). We then evaluate how well these generated questions align with reference questions from the patient's original profile.

We construct simulated patients from real patient posts on online forums, treating the questions that patients ultimately asked in their posts as golden reference questions. We hypothesize that patients who are able to make detailed posts are well-informed, and the questions they formulate represent well-prepared, high-quality questions. We clean and process these questions to serve as reference questions for evaluation.

### 5.2 Metrics

We evaluate question quality using ROUGE-1, ROUGE-L, and BLEU-1, BLEU-2. Note we considered using the same cognitive load metrics from the human study with simulated patients but pursued this reference-based approach; see Sec 6 for our lessons. Since generated questions and references are both multiple, we use macro-oracle evaluation (Sai et al., 2020; Liu and Liu, 2021; Ravaut et al., 2022): for each input with $K$ generated candidates and $M$ references, we compute ROUGE for all $(h_i, r_j)$ pairs, take the maximum over candidates for each reference, then average across references: $\frac{1}{M}\sum_j \max_i \text{ROUGE}(h_i, r_j)$.

We evaluate against two types of reference questions: (1) *original questions*, which are questions as originally posted by patients, often colloquial and incomplete (e.g., "Why waiting for incontinence to improve?"), and (2) *refined questions*, which are professionally refined versions that are clearer and more complete (e.g., "Why does my doctor recommend waiting until my urinary incontinence improves to one pad per day before starting radiation?"), generated using GPT-5 to improve clarity while preserving the original intent.

### 5.3 Simulated Patient Construction

**Collection for Profiles and Reference Questions.** We construct simulated patients from real patient posts on Reddit's r/ProstateCancer forum. We crawl posts where patients discuss their prostate cancer experiences, concerns, and questions. For each post, we extract questions using a structured prompt that identifies patient questions, refines them for clarity and completeness, and categorizes them by type (e.g., Treatment Plan, Long-Term Management). We preserve the surrounding context from the original post for extracted questions as patient profile information. To ensure multiple references for robust evaluation, we filter posts that contain only one question. We also filter posts where questions are unrelated to the post content or seek peer experiences. By tracing three months of posts on the forum, we successfully construct 43 simulated patient profiles from this process. As shown in Tab 2, most profiles correspond with 2-3 questions.

| # Profiles | Avg # Q/Profile | Distribution (# Profiles) 2 Q | 3 Q | 4 Q | 6 Q |
|---|---|---|---|---|---|
| 43 | 2.56 | 27 | 10 | 5 | 1 |

Table 2: Simulated Patient Profiles and Question Distribution

**Patient System.** Following prior work (Li et al., 2024) that has successfully built patient systems for medical benchmarks, we convert the context from each post into a set of atomic facts—discrete, verifiable statements about the patient's situation, which we organize into structured patient profiles. These atomic facts serve as constraints during patient role-play, ensuring that simulated patient responses are grounded in the original post content and reducing hallucination. During role-play, each simulated patient responds to interview questions using only information from their atomic facts, ensuring responses remain faithful to the original post content while allowing natural conversation exploration.

### 5.4 Results

Tab 3 presents scores comparing the plain LLM and our system from simulated patients. Our system shows consistent improvements over the plain LLM, particularly when evaluated against refined reference questions. ROUGE-1 increases from 23.1 to 25.0 ($p < 0.05$), ROUGE-L increases from 18.1 to 20.3 ($p < 0.01$), and BLEU-1 increases from 14.0 to 15.6 ($p < 0.05$). When evaluated against original questions, these differences are not statistically significant but still show modest improvements. The colloquial nature of original questions and the sample size likely contribute to the lack of statistical significance. Together with the observations from the human patient study, these results suggest that our system generates questions that better align with patient information-seeking needs, particularly when evaluated against professionally refined reference questions. The framework-guided interview structure and retrieval-augmented generation appear to help the system generate more relevant and comprehensive questions for patients to bring to their medical visits.

## 6 Lessons & Conclusion

In this work, we explore designs for building a conversational agent to help patients prepare for their upcoming medical visits. Our evaluation demonstrates that the framework-guided interview structure and retrieval-augmented generation bring benefits, generating more relevant questions and reducing cognitive load compared to a plain LLM. Through our study, we also draw several lessons for future work.

| Metric | Refined Questions: Plain LLM | Refined Questions: Ours | Original Questions: Plain LLM | Original Questions: Ours |
|---|---|---|---|---|
| ROUGE-1 | 23.1 | **25.0**$^{*}$ | 15.3 | **16.2** |
| ROUGE-L | 18.1 | **20.3**$^{**}$ | 13.2 | **13.8** |
| BLEU-1 | 14.0 | **15.6**$^{*}$ | 8.3 | **8.5** |
| BLEU-2 | 5.6 | **7.0** | 2.7 | **2.8** |

Table 3: ROUGE and BLEU scores for generated questions. $^{*}p < 0.05$, $^{**}p < 0.01$ (paired t-test/Wilcoxon test, $n = 43$)

**Balancing Responsiveness and Framework Control.** In early iterations, we fully structured the interview with template questions. Pilot study participants found the system unresponsive and frustratingly rigid. We improved by treating template questions as a backbone to guide the flow, allowing the LLM to personalize responses while maintaining clinical structure.

**Challenges with Simulated Cognitive Load Drive Reference-Based Evaluation.** We attempted to use the same cognitive load assessment with simulated patients, but their ratings were highly homogeneous with no difference between variants. Their perceptions also differ: simulated patients feel burdened by more turns with focused questions, while humans find plain LLM's long messages with many questions per turn more burdensome. This led us to anchor real-world data as reference for meaningful evaluation.

**Expanding Retrieval Sources Will Improve User Experience While Maintaining Clinical Reliability.** Some feedback noted that while patients trust our system, the plain LLM seems to have a larger knowledge base, answering questions beyond our designated guidebook (e.g., locating local recovery communities). Our retrieval sources were limited to a general guidebook. Expanding to include more validated books and websites could help compete with plain LLMs while maintaining clinical reliability.

**More Steps Needed Before Real-World Deployment.** Participants also raised ethical concerns in feedback forms. We identify several limitations of the current system in terms of safety and discuss them in the Limitations section (Section 6).

## Ethical Considerations

This study received IRB approval from our institution. All participants provided informed consent and were informed of potential risks including technical errors, hallucinations, and incomplete information. We ensured privacy by collecting no personally identifiable information beyond age and race/ethnicity. User interactions were logged securely. Participants were compensated at $20 per hour per Prolific platform standards.

While these safeguards were appropriate for our pilot study, this system represents early-stage research not intended for clinical use without substantial further development and validation. As detailed in the Limitations section, future work must include improved safety mechanisms, transparent risk communication, clinical integration protocols, and formal randomized controlled trials before this research prototype can be considered a clinical tool.

## Limitations

This study has several limitations requiring resolution before real-world deployment. Our evaluation used participants with localized prostate cancer experience simulating hypothetical visit preparation rather than actual scheduled appointments. This may not capture the emotional weight, urgency, and decision-making pressures of actual medical visits. Additionally, focus on localized prostate cancer with small sample size (n=10) limits generalizability to other cancer types and populations, requiring adaptation and validation for different oncological contexts.

To facilitate safe and effective real-world application, several improvements are needed:

1. **Enhanced Safety Controls:** More robust safeguards to detect and mitigate harmful or inaccurate outputs, flag uncertain information, and redirect users to verified medical resources.

2. **Transparent Risk Communication:** Explicit disclosure of AI hallucination risks and possibility of incorrect or incomplete information.

3. **Clinical Integration:** Generate summary cards for patients to bring to appointments, alerting physicians to AI-assisted preparation and enabling them to address misconceptions or gaps.

4. **Improved Interview Proactiveness:** Enhance the interview to more actively probe concerns, identify knowledge gaps, and adapt questioning based on responses.

Finally, randomized controlled trial (RCT) evaluation is essential to establish true efficacy. Such studies should compare AI-assisted preparation against human medical assistants or traditional educational resources, measuring: (1) time to visit-readiness, (2) patient confidence levels, (3) question quality and clinical appropriateness, and (4) physician assessment of patient preparedness during visits. While logistically challenging, this validation is necessary to justify integration into cancer care workflows.

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| Role | Content |
|---|---|
| Assistant | Hello. I am your Prostate Cancer Patient Assistant. I'm here to help you prepare for your upcoming medical visit. To do that, I'd like to understand what choices or uncertainties you're currently facing in your prostate cancer journey. This will help ensure you get the most out of your conversation with your doctor. Are you ready to get started? |
| Patient | Yes. Let's get started |
| Assistant | First, we would like to know whether you currently have any concerns, questions, or decisions related to one of the optional areas. Please select the category of concerns you're currently thinking about as part of your decision-making. |
| Patient | Long-Term Management and Monitoring |
| Assistant | It's great to hear that you're thinking about long-term management and monitoring, as it's an important part of managing prostate cancer effectively. Understanding your plan can help you feel more in control and prepared for the future. What do you understand about your follow-up care or monitoring plan? Is there any specific aspect of your long-term management that you're particularly concerned about or want more information on? |
| Patient | From what I understand, my follow-up care will mostly involve regular PSA tests and maybe occasional imaging... |
| Assistant | It sounds like you're doing a great job of staying informed about your follow-up care, which is crucial for long-term management. Balancing this with arthritis can be challenging, so prioritizing self-care and understanding warning signs is key. How do you currently manage your arthritis symptoms, and have you discussed any integrated care approaches with your doctor? Do you have any questions or concerns about how you'll manage this condition in the future? What are they? |
| Patient | Right now, I manage my arthritis mostly with pain medication and trying to stay active when I can, though ... |
| Assistant | It sounds like you're handling a lot, and it's understandable to feel concerned about managing both arthritis and prostate cancer. It's important to remember that integrating care for both conditions can help you maintain a balanced lifestyle and reduce stress. Have you ever considered or tried any non-medication strategies, such as physical therapy or lifestyle changes, that might help manage both your arthritis and prostate health? |
| Patient | I've thought about it, but I haven't really tried anything structured like physical therapy yet... |
| Assistant | It's great that you're thinking about incorporating structured approaches like physical therapy into your routine, as they can offer significant benefits for managing both arthritis and prostate health. Having a tailored plan can make it easier to stay consistent and feel more in control. Have you ever discussed with a healthcare professional the possibility of a tailored exercise program or received advice on specific lifestyle changes that could benefit both conditions? |
| Patient | No, I haven't really talked with a healthcare professional about a specific exercise program or lifestyle changes that ... |
| Assistant | **Knowledge Panel**<br>Thanks for your sharing. Based on what you've shared, we've identified the following information that may help you make a decision about **long-term management and monitoring** or address your concerns in this area.<br>**Knowledge:** Active surveillance is a common approach for managing low-risk prostate cancer, allowing patients to maintain quality of life while monitoring for any signs of progression. Regular PSA tests, digital rectal exams, and periodic biopsies or MRIs are typical components of this strategy. It's important to discuss with your doctor the frequency of these tests and any lifestyle changes that could support your health during this period.<br>**Key factors:** (1) Balancing prostate cancer monitoring with arthritis management; (2) Understanding the frequency and type of tests required for active surveillance; (3) Exploring integrated care approaches for managing both prostate cancer and arthritis.<br>**Options:** *Active Surveillance* – Benefits: Maintains quality of life, delays potential side effects. Risks: Requires regular monitoring, potential anxiety. *Surgery or Radiation Therapy* – Benefits: Potentially curative, immediate treatment. Risks: Possible side effects impacting quality of life. *Integrated Care Approach* – Benefits: Holistic management of both conditions, potentially improving overall well-being. Risks: Requires coordination among healthcare providers, may involve lifestyle adjustments.<br>*For your reference: This information is based on the Patient Guide to Localized Prostate Cancer (Page 18, Page 29-30, Page 41, Page 45).* |
| Patient | I have read it. |
| Assistant | Having professional guidance on a tailored exercise program could make a significant difference in managing both your arthritis and prostate health, especially considering your current challenges with consistency due to pain. It may be beneficial to bring this up with your healthcare team to explore options that fit your needs. Which options do you currently see as available to you? Please list all of them out. |
| Patient | At the moment, the ways I'm managing both my arthritis and prostate health are mostly using my pain meds for arthritis ... |
| Assistant | It's commendable that you are actively managing your symptoms and considering professional guidance to improve your control over both conditions. Discussing a tailored plan with your healthcare provider could ensure you have a comprehensive strategy that accommodates your arthritis while keeping up with prostate cancer monitoring. Have you thought about setting specific goals with your doctor to address both conditions together? What parts still feel unclear? |
| Patient | I've thought about setting goals with my doctor, but we haven't really done that yet... |

Table 4: Example conversation between the system and a simulated patient (Part 1 of 2). Template questions are highlighted in blue.

| Role | Content |
|---|---|
| Assistant | It's great that you're considering setting specific goals with your doctor, as this can provide a clearer roadmap for managing both your arthritis and prostate health. Having a structured plan can help you feel more confident and in control. Which factors matter most to you in this decision? (If possible, list your top 2–3.) You can also propose your own factors. Balancing prostate cancer monitoring with arthritis management. Understanding the frequency and type of tests required for active surveillance. Exploring integrated care approaches for managing both prostate cancer and arthritis. |
| Patient | What matters most to me is finding a balance between keeping up with prostate cancer monitoring and managing ... |
| Assistant | It sounds like you're focused on achieving a balance that allows you to manage both your prostate cancer monitoring and arthritis effectively. Understanding the necessary frequency and types of tests can certainly help alleviate some stress, making it easier to commit to your health plan. Which tradeoff feels hardest to weigh right now? |
| Patient | The hardest part for me right now is figuring out how to stay active with my arthritis while also keeping up with ... |
| Assistant | Finding the right balance between staying active with arthritis and keeping up with prostate cancer monitoring is indeed a challenging task, especially when dealing with pain and stiffness. It might be helpful to discuss with your healthcare team strategies that can accommodate both needs, perhaps through modified exercises or a flexible monitoring schedule that considers your arthritis flare-ups. What makes you feel less confident or ready? |
| Patient | I think what makes me feel unprepared is not really knowing what I can safely do with my arthritis while still keeping ... |
| Assistant | It's understandable to feel uncertain about how much activity is right for you, especially when managing two conditions. Your healthcare team can provide guidance on exercises that accommodate both your arthritis and prostate cancer needs, helping you feel more confident in your routine. Have you discussed this with anyone (family, friends, clinicians, support groups)? What did they say? |
| Patient | Not really, I haven't talked much about it with anyone yet... |
| Assistant | It sounds like you're doing your best to navigate these challenges on your own, and reaching out to your family is a great first step. Joining a support group or discussing your concerns with your healthcare team might provide additional insights and encouragement. What steps do you think you could take to start involving your healthcare team in creating a plan that addresses both your arthritis and prostate cancer needs? |
| Patient | I think the first step would be to have a proper conversation with my doctor where I can explain all my concerns ... |
| Assistant | It's a great idea to bring a list of questions and track your symptoms; this can provide your doctor with a clearer picture of your needs. Collaborating with a physical therapist or specialist could offer tailored strategies that align with both your arthritis and prostate cancer management. How comfortable do you feel about initiating these discussions with your healthcare team, and is there anything specific you might need to help you prepare for that conversation? |
| Patient | I feel a little nervous about bringing it up because I don't want to take up too much time or forget something important ... |
| Assistant | Thank you for your time today. We have received all your responses. This interview has been completed. |
| Assistant | **Know-Them Questions:**<br>*Educational background: The answers below are from Patient Guide to Localized Prostate Cancer. Not medical advice—please confirm with your clinician.*<br>• **What type of testing will be required if I do active surveillance?** Men on active surveillance will usually have a PSA blood test done twice per year, an annual DRE, and repeat biopsies generally every 1 to 5 years. MRI may also be used to help monitor the cancer and guide timing and decision for repeat biopsies.<br>• **How frequently will I be tested under active surveillance?** PSA testing is done twice per year, and MRIs are generally performed every two years to monitor any changes in the prostate.<br>• **What are proactive changes to my health habits that I should consider?** Exercise as regularly as you can, most days of the week, and increase the intensity of your exercise as you are able. An anti-inflammatory, heart-healthy diet is also recommended.<br>• **What is my baseline PSA number, and what number would be concerning?** Questions about the specific tests needed under active surveillance and PSA benchmarks are important to discuss with your doctor.<br>• **How does exercise impact prostate cancer recovery and long-term health?** Exercise reduces risk of prostate cancer recurrence and improves survival. It also supports better sleep, cognition, and reduces depression and anxiety.<br>**Ask-Them Questions:**<br>• How can I safely integrate exercises that benefit both my arthritis and prostate health?<br>• What specific lifestyle changes should I make to manage both prostate cancer and arthritis effectively?<br>• Are there any specific warning signs I should watch for at home that indicate a change in my prostate cancer status?<br>• How can I balance the need for physical activity with the limitations imposed by my arthritis?<br>• Would involving a physical therapist be beneficial for creating a tailored exercise program for my conditions? |

Table 5: Example conversation between the system and a simulated patient (Part 2 of 2). Template questions are highlighted in blue.